

Exercise 11

AIM :

To create a C program for below logic

- 1.Create a functions to create new linked list
- 2.add list at the front
- 3.add list at the back
- 4.Print all the data in the list

ALGORITHM :

Step 1: Start the Program

1. Begin execution of the program.

Step 2: Define the Node Structure

2. Define a structure Node containing:
 - An integer data
 - A pointer next to the next node

Step 3: Declare Function Prototypes

3. Declare functions for:
 - Creating a new linked list
 - Inserting a node at the front
 - Inserting a node at the back
 - Displaying the linked list
 - Freeing allocated memory

Step 4: Initialize Head Pointer

4. Initialize a pointer head to NULL to represent an empty linked list.

Step 5: Display Menu in a Loop

5. Repeat the following steps until the user chooses to exit:
 - Display a menu of operations:
 1. Create new list

2. Add node at front
3. Add node at back
4. Print list
5. Exit

- Read the user's choice.

Step 6: Perform Operation Based on Choice

Case 1: Create New List

6. If a list already exists, free all nodes.
7. Read the initial data value.
8. Allocate memory for a new node.
9. Store the data and set next = NULL.
10. Assign the new node as the head.

Case 2: Add Node at Front

11. If the list is empty, display an error message.
12. Otherwise:
 - Allocate memory for a new node.
 - Store the data.
 - Set newNode->next = head.
 - Update head to the new node.

Case 3: Add Node at Back

13. If the list is empty, display an error message.
14. Otherwise:
 - Allocate memory for a new node.
 - Store the data and set next = NULL.
 - Traverse the list until the last node.
 - Set lastNode->next = newNode.

Case 4: Print the Linked List

15. If the list is empty, display a message.
16. Otherwise:
 - Traverse from head to NULL.

- Print each node's data in sequence.

Case 5: Exit

17. Free all dynamically allocated nodes.
18. Terminate the program.

Default Case

19. Display an error message for invalid input.

Step 7: Stop the Program

20. End execution.

OUTPUT :

```
PS D:\C programming\DHAVAMANI_EX11> gcc main.c -o main
PS D:\C programming\DHAVAMANI_EX11> ./main
```

Choose operation:

1. Create new list
2. Add node at front
3. Add node at back
4. Print list
5. Exit

Enter your choice: 2

List is empty! Create a list first.

Choose operation:

1. Create new list
2. Add node at front
3. Add node at back
4. Print list
5. Exit

Enter your choice: 1

Enter initial data: 55

Created new list with initial data: 55

Choose operation:

1. Create new list
2. Add node at front
3. Add node at back
4. Print list
5. Exit

Enter your choice: 2

Enter data to add at front: 45

Added 45 at the front

```
Choose operation:
1. Create new list
2. Add node at front
3. Add node at back
4. Print list
5. Exit
Enter your choice: 3
Enter data to add at back: 65
Added 65 at the back
```

```
Choose operation:
1. Create new list
2. Add node at front
3. Add node at back
4. Print list
5. Exit
Enter your choice: 4
Linked List: 45 -> 55 -> 65
```

```
Choose operation:
1. Create new list
2. Add node at front
3. Add node at back
4. Print list
5. Exit
Enter your choice: 5
Exiting...
```